MILITARY SPECIFICATION

COUPLING ASSEMBLY, QUICK DISCONNECT, PLAIN AND SELF-SEALING

1. SCOPE

1.1 Scope. This specification covers three types of quick disconnect couplings.

1.2 Classification. Couplings shall be of the following types (see drawings for class and style):

Type I — Plain.
Type II — Self-Sealing.
Type III — Combination (type I and type II).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL–P–5315 — Packing, "O" Ring Hydrocarbon Fuel Resistant.

MIL–H–5606 — Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.

MIL–P–25732 — Packing, Preformed, Petroleum Hydraulic Fluid Resistant, 275 Deg. F.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–130 — Identification Marking of U. S. Military Property.

MIL–STD–171 — Finishing of Metal and Wood Surfaces.

DRAWINGS

EDGEWOOD ARSENAL

C150–1–1 — Coupling Half, Quick Disconnect, Nipple, Type I—Plain Size 4, 1/4 Nominal Tube OD.

C150–1–2 — Coupling Half, Quick Disconnect, Nipple, Type I—Plain Size 6, 3/8 Nominal Tube OD.

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C150-1-3 — Coupling Half, Quick Disconnect, Nipple, Type I—Plain Size 8, ½ Nominal Tube OD.

C150-1-4 — Coupling Half, Quick Disconnect, Nipple, Type I—Plain Size 12, ¾ Nominal Tube OD.

C150-1-5 — Coupling Half, Quick Disconnect, Nipple, Type I—Plain Size 16, 1 Nominal Tube OD.

C150-1-6 — Coupling Half, Quick Disconnect, Nipple, Type I—Plain Size 20, 1¼ Nominal Tube OD.

C150-1-7 — Coupling Half, Quick Disconnect, Nipple, Type I—Plain Size 24, 1½ Nominal Tube OD.

C150-1-8 — Coupling Half, Quick Disconnect, Nipple, Type I—Plain Size 32, 2 Nominal Tube OD.

C150-1-9 — Coupling Half, Quick Disconnect, Nipple, Type I—Plain Size 40, 2½ Nominal Tube OD.

C150-1-10 — Coupling Half, Quick Disconnect, Nipple, Type I—Plain Size 48, 3 Nominal Tube OD.

D150-1-11 — Coupling Half, Quick Disconnect, Coupling, Type I—Plain.

D150-1-12 — Coupling Half, Quick Disconnect, Coupling or Nipple, Type II, Self-Sealing.

D150-1-13 — Cap, Protective.

D150-1-14 — Cap, Protective and Pressure.

D150-1-15 — Plug, Protective.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification rules should addressed to the Uniform Classification Committee; 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 *Materials.* All materials shall conform to the specifications and characteristics set forth on the applicable drawing. Metals not corrosion resistant shall be treated to resist corrosion. Dissimilar metals shall not be used in intimate contact with each other unless they are protected against electrolytic corrosion. Dissimilar metals are defined in MIL-STD-171.

3.1.2 *Components.* All components of the couplings shall conform to the specifications and applicable drawing.

3.2 Assembly.

3.2.1 *Type I—plain coupling.* The coupling

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shall be assembled in accordance with Drawing D50–1–11.

3.2.2 *Type II—self-sealing.* The coupling shall be assembled in accordance with Drawing D150–1–12.

3.3 Nipples.

3.4.1 *Type I.* The plain couplings (see accordance with the applicable drawing specified in table I.

TABLE I. *Drawings*

Size	Drawing
¼	C150–1–1
⅜	C150–1–2
½	C150–1–3
¾	C150–1–4
1	C150–1–5
1¼	C150–1–6
1½	C150–1–7
2	C150–1–8
2½	C150–1–9
3	C150–1–10

3.3.2 *Type II—self-sealing.* The nipples shall be in accordance with Drawing D150–1–12.

3.4 Interchangeability.

3.4.1 *Type I.* The plain couplings (see 3.2.1) shall mate with the plain nipples (see 3.3.1) and all couplings of like size shall be interchangeable.

3.4.2 *Type II.* The self-sealing couplings (see 3.2.2) shall mate with the self-sealing nipples (see 3.3.2) and all couplings of like size shall be interchangeable.

3.4.3 *Type III.* The self-sealing coupling (see 3.2.2) shall mate with the plain nipple (see 3.3.1) and the valve shall be actuated by the plain nipple.

3.5 Seals. All couplings shall be equipped with seals made from material conforming to MIL–P–5315 or MIL–P–25732.

3.6 Functioning.

3.6.1 *Low temperature.* The sample couplings and nipples shall be subjected to a temperature of —65° ± 5°F for 72 hours and then connected and disconnected 25 times. Their ability to function or assemble shall not be affected when tested as specified in 4.6.1.1.

3.6.2 *High temperature.* The sample couplings and nipples shall be subjected to a temperature of 275° ± 5°F for 72 hours at a relative humidity of 75 percent and then connected and disconnected 25 times. Their ability to function or assemble shall not be affected when tested as specified in 4.6.1.1.

3.6.3 *Valve spillage.* The self-sealing coupling and nipple when subjected to a static pressure of 10 pounds per square inch gage (psig) and quickly disconnected shall not leak more fluid than specified in table II when tested as specified in 4.6.1.2.

TABLE II. *Spillage on disconnect*

Coupling size (inches)	Spillage (milliliters)
¼	1.2
⅜	1.4
½	4.2
¾	10.5
1	19.6
1¼	30.8
1½	55.2
2	85.3
2½	172.0
3	262.0

3.7 Leakage. The coupling assemblies shall not leak when subjected to the static pressures in accordance with table III when tested as specified in 4.6.2.

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TABLE III. *Maximum pressure (psig)*

Specified material	Size									
	¼	⅜	½	¾	1	1¼	1½	2	2½	3
Aluminum	4000	4000	3500	3500	3000	750	1000	600	600	400
Brass	4000	4000	4000	4000	3500	700	700	800	800	400
Stainless steel	10000	8000	7500	7000	4000	3000	3000	1000	800	800
Steel	13000	9000	8000	7000	4000	3500	3000	3000	2000	1500

3.8 Working pressure. The sample coupling assemblies shall not leak and shall withstand 100,000 cycles at —65° ± 5° F and 100,000 cycles at 165° ± 5° from 0 psig to the pressure in accordance with table IV and tested as specified in 4.6.3.

TABLE IV. *Working pressure (psig)*

Specified material	Size									
	¼	⅜	½	¾	1	1¼	1½	2	2½	3
Aluminum	2250	2250	1750	1750	1500	375	375	300	300	200
Brass	2250	2250	2000	2000	1750	350	350	400	400	200
Stainless steel	5000	4000	3750	3500	2000	1500	1500	500	400	400
Steel	6500	4500	4000	3500	2000	1750	1500	1500	1000	750

3.9 Pressure drop. The coupling assemblies when subjected to the flow rate in gallons per minute in accordance with table V shall show no more than 0.6 psig pressure drop when tested as specified in 4.6.4.

TABLE V. *Test flow rates*

Type	Size									
	¼	⅜	½	¾	1	1¼	1½	2	2½	3
Type I	2	4	5.5	10	20	26	42	82	130	160
Type II	0.75	1.4	1.6	3.8	6.5	10	14	26	38	50
Type III	0.9	1.8	2.3	5	8	14	20	38	55	73

3.10 Connect force. The coupling assemblies when subjected to the pressure in accordance with table VI shall connect at the force listed in table VI when tested as specified in 4.6.5.

TABLE VI. *Connect force*

Size	¼	⅜	½	¾	1	1¼	1½	2	2½	3
Internal pressure (psig).	200	200	200	100	100	100	50	50	50	50
Force to connect (lb).	50	60	95	95	145	195	155	245	425	560

3.11 Leakage. The self-sealing coupling half (disconnected) shall not leak when subjected to the pressures in accordance with table VII and tested as specified in 4.6.6.

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TABLE VII. *Static pressure*

Size	¼	⅜	½	¾	1	1¼	1½	2	2½	3
Internal pressure (psig).	200	200	200	100	100	100	50	50	50	50

3.12 Connection. For safety the self-sealing couplings shall function only when completely connected or disconnected.

3.13 Maintenance. The couplings shall be designed to permit overhaul, repair or removal of seals without special tools.

3.14 Protection. The couplings when installed and in service may be protected by the protective cap or plug, Drawing D150–1–13, D150–1–14 and D150–1–15 (see 6.2(d)).

3.15 Identification. All coupling assemblies of nominal size of 1 inch or more shall be marked in accordance with MIL–STD–130.

3.16 Preproduction. When specified by the contract or order, a preproduction sample of couplings shall be produced for examination and test in accordance with this specification.

3.17 Workmanship. The coupling assemblies shall be free from foreign matter (oil, grease or dirt) and damage such as dents, cracks or deformation which would affect their intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a writen proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2, 3.3, 3.4, 3.5, 3.12, 3.13, 3.14 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 *Inspection substations.* If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL–STD–105 are applicable to each substation and to the end item.

4.3 Lotting. A lot shall consist of the couplings produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.4 Sampling.

4.4.1 *For examination.* Sampling shall be conducted in accordance with MIL–STD–105.

4.4.2 *For tests.* Sampling shall be conducted in accordance with MIL–STD–105, level S–4.

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4.5 Inspection provisions.

4.5.1 *For examination.* Sample couplings shall be examined and tested in accordance with the classification of defects and MIL–STD–105.

4.5.2 *For tests.* Sample components shall be tested in accordance with 4.6 and MIL–STD–105 using an AQL of 0.65 percent defective.

4.5.3 *Classification of defects.*

4.5.3.1 *Couplings,* quick disconnect.

Categories	*Defects*
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Identification missing or incorrect
102	Seal missing
103	Workmanship (3.17)

4.6 Tests.

4.6.1 *Functioning.*

4.6.1.1 *Low and high temperature.* The nipple shall be disconnected from the coupling and the halves shall be conditioned in the required environmental conditions for a minimum of 72 hours. After conditioning, the halves shall be connected and disconnected for the specified number of times (use gloves) without the use of tools.

4.6.1.2 *Valve spillage.* The coupling end shall be connected to the high pressure side of a pump and hydraulic oil conforming to MIL–H–5606 shall be pumped and circulated. The coupling shall be disconnected and all drippings shall be caught.

4.6.2 *Leakage.* The coupling assembly shall be subjected to the static hydraulic pressure as specified in table III for 5 to 6 minutes to determine whether the coupling assembly leaks.

4.6.3 *Working pressure.* The coupling assembly shall be subjected to a static pressure of 10 psig and conditioned in the required environmental conditions for 24 hours. After conditioning, the pressure in the coupling shall be oscillated from ambient to the pressure specified in table IV at the rate of 35 ± 5 cycles per minute for the specified number of cycles.

4.6.4 *Pressure drop.* The pressure drop shall be determined by measuring the drop in pressure across the coupling with a calibrated manometer using the flow rates specified in table V.

4.6.5 *Connect force.* A vise-like fixture equipped with a spring scale which measures the force exerted by the jaws of the vise to the nearest ½ pound shall be used. The pressure specified in table VI shall be applied in the coupling half. The jaws of the vise shall be closed, the coupling sleeve shall be pulled back and the force at which the coupling connects shall be determined.

4.6.6 *Leakage.* Pressure shall be applied to the separated halves of the coupling assembly at a rate of 3 pounds per square inch per minute until the pressure specified in table VII has been attained. (The pressure may be applied continuously or in 3 pound intervals. If the pressure is applied in 3 pound intervals, one minute interruptions should be allowed between intervals.) Leakage of the valves during this pressurization period should be determined.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The coupling assemblies shall be packaged to afford adequate protection from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Coupling assemblies, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall comply with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

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5.3 **Marking.** In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 **Intended use.** These couplings are intended for use in high pressure air or hydrocarbon fuel applications.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Type, size, class and end style required.

(c) Preproduction (when required).

(1) Quantity required.

(2) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(3) Name and address of test facility when testing is performed by the Government.

(4) Packing and shipping instructions when applicable.

(5) Time required for the Government to notify the supplier whether or not to proceed with production.

(d) The need for protective cap or plug.

6.3 **Supersession data.** This specification supersedes Chemical Corps Purchase Description 197–54–823, dated 3 April 1962.

Custodian:

Army—MU

Preparing activity:

Army—MU (EA)

Project No. 4730–A127

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1426 REPLACES NAVSHIPS FORM [illegible], [illegible] WHICH ARE OBSOLETE

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